EE2015: Electric Circuits and Networks

Tutorial 4 Answer Key
(September 12, 2025)

1. $\mathbf{Z}_{ab} = 7.567 + j0.5946\,\Omega$

   $\mathbf{Z}_{ab} = 0.3796 + j1.46\,\Omega$

2. $\mathbf{Z} = \begin{bmatrix} 2.7333 & 0.0667 \\ 0.0667 & 2.7333 \end{bmatrix}$

   $\mathbf{Y} = \begin{bmatrix} 0.3661 & -0.0089 \\ -0.0089 & 0.3661 \end{bmatrix}$

3. $\mathbf{Z} = \begin{bmatrix} 9.6 & -0.8 \\ -0.8 & 8.4 \end{bmatrix}$

   $\mathbf{Y} = \begin{bmatrix} 0.105 & 0.01 \\ 0.01 & 0.12 \end{bmatrix}$

4. $\mathbf{Z}(s) = \begin{bmatrix} R_1 + R_3 + s(L_1 + L_3 + 2M_1) & R_3 + s(L_3 - M_2 + M_1) \\ R_3 + s(L_3 - M_2 + M_1) & R_2 + R_3 + s(L_2 + L_3) \end{bmatrix}$

5. $\mathbf{Z} = \begin{bmatrix} 25 & 20 \\ 24 & 30 \end{bmatrix}$